When Are Semantic Networks Hyperbolic? A Curvature Phase Transition Theory with Cross-Linguistic Validation

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Abstract

Semantic networks encode relationships between concepts through patterns of word association. Whether these networks exhibit hyperbolic geometry—and why—remains an open question with implications for cognitive architecture and network embedding. We present a unified theory that combines (i) a curvature phase transition in random graphs with (ii) empirical analysis of 11 semantic networks across five languages and one clinical dataset.

Using exact linear programming to compute Ollivier–Ricci curvature (eliminating Sinkhorn regularization bias), we find that the density parameter $\eta = \langle k \rangle^2/N$ is necessary but not sufficient for predicting hyperbolicity. Random k-regular graphs undergo a sign change in mean curvature $\bar{\kappa}$ at a critical density $\eta_c(N) = 3.75 - 14.62/\sqrt{N}$ ($R^2 = 0.995$). All semantic networks except Dutch SWOW fall below this threshold ($\eta \ll \eta_c$), yet taxonomies (WordNet, BabelNet) are near-Euclidean while association networks (SWOW, ConceptNet) are hyperbolic. The missing parameter is clustering coefficient C: networks with $C > 0.10$ are hyperbolic ($\bar{\kappa} \in [-0.24, -0.07]$), those with $C < 0.02$ are Euclidean ($\bar{\kappa} \approx 0$), and those with $\eta > \eta_c$ are spherical ($\bar{\kappa} > 0$).

Dutch SWOW ($\eta = 7.56 \gg \eta_c = 3.10$) is spherical ($\bar{\kappa} = +0.10$), confirming the phase transition prediction in the opposite direction. Sphere-embedded (S^3) transport cost flips 7 of 9 networks from negative to positive curvature, demonstrating that semantic hyperbolicity is largely metric-dependent. A Lean 4 formalization (25 modules, 0 `sorry` in 7 core ORC-theory modules) provides machine-checked proofs of Wasserstein non-negativity, curvature bounds, and regime exclusivity.

Keywords: semantic networks, Ollivier–Ricci curvature, phase transition, hyperbolic geometry, clustering coefficient, cross-linguistic, formal verification

1 Introduction

1.1 Background

Semantic memory—the structured knowledge of concepts and their relationships—is fundamental to human cognition. Network science offers a quantitative lens on this organization, treating words as nodes and associations as edges to uncover small-world structure, modularity, and degree heterogeneity [2]. Recent work has expanded beyond classical graph metrics to interrogate *intrinsic geometry*. Hyperbolic spaces, in particular, naturally accommodate hierarchical growth and heterogeneous branching—properties observed in semantic association datasets [3]. Yet most evidence for hyperbolicity in language has been indirect, relying on embeddings or curvature heuristics rather than explicit geometric measurement.

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1.2 The Open Question

Two questions have remained separate in the literature:

1. **Empirical**: Do semantic networks exhibit hyperbolic geometry? Is this universal across languages?
2. **Theoretical**: Why should some networks be hyperbolic and not others? What are the boundary conditions?

Prior work addressed these independently. Empirical studies applied Ollivier–Ricci curvature to specific datasets and found hyperbolicity in some but not all networks [8, 11]. Theoretical studies established sign changes in discrete curvature for random graph models [15]. No work has connected the two: using the theoretical phase transition to *predict* which real networks should be hyperbolic.

1.3 Our Contribution

We present a unified theory that bridges random graph phase transitions with empirical semantic network analysis:

1. **Two-parameter predictive theory**: Hyperbolicity requires both (a) $\eta = \langle k \rangle^2/N < \eta_c(N)$, and (b) sufficient clustering ($C \gtrsim 0.10$). Neither alone is sufficient.
2. **Exact LP computation**: Ollivier–Ricci curvature for 11 networks via exact linear programming (JuMP + HiGHS), eliminating Sinkhorn bias.
3. **Cross-linguistic validation**: Networks spanning Spanish, English, Chinese, Dutch, Portuguese, Russian, and Arabic.
4. **Dutch SWOW confirmation**: The dense Dutch SWOW ($\eta = 7.56 > \eta_c$) is spherical ($\bar{\kappa} = +0.10$), confirming the phase transition.
5. **Metric dependence**: Sphere-embedded ORC flips most networks from hyperbolic to spherical.
6. **Lean 4 formalization**: 25 modules with 0 `sorry` in 7 core ORC-theory modules.

2 Theory: Phase Transition in Discrete Curvature

2.1 Ollivier–Ricci Curvature

For an edge (u, v) in a graph G:

$$\kappa(u, v) = 1 - \frac{W_1(\mu_u, \mu_v)}{d(u, v)} \tag{1}$$

where $\mu_u = \alpha\, \delta_u + (1-\alpha) \cdot \mathrm{Uniform}(N(u))$ is the probability measure with idleness $\alpha = 0.5$, and W_1 is the Wasserstein-1 distance [1]. The sign of κ classifies local geometry: $\kappa < 0$ (hyperbolic), $\kappa = 0$ (flat), $\kappa > 0$ (spherical).

We compute W_1 via exact linear programming [5]:

$$\min \sum_{i,j} d_{ij}\, \gamma_{ij} \quad \text{s.t.} \quad \sum_j \gamma_{ij} = \mu_i, \quad \sum_i \gamma_{ij} = \nu_j, \quad \gamma_{ij} \geq 0 \tag{2}$$

using the HiGHS solver [6]. This eliminates the entropy regularization bias of the Sinkhorn approximation [4] (quantified as $|\Delta\kappa| < 0.015$ for random graphs).

2.2 Phase Transition in Random Regular Graphs

For random k-regular graphs on N vertices, the mean curvature $\bar{\kappa}$ undergoes a sign change at a critical density η_c where $\eta = k^2/N$:

- **Sparse regime** ($\eta < \eta_c$): $\bar{\kappa} < 0$ (hyperbolic). Tree-like local structure dominates.
- **Dense regime** ($\eta > \eta_c$): $\bar{\kappa} > 0$ (spherical). Triangle-rich structure dominates.

Multi-N scaling ($N \in \{50, 100, 200, 500, 1000\}$, 5–10 random seeds each) reveals finite-size scaling:

$$\eta_c(N) = \eta_c^\infty - \frac{a}{\sqrt{N}}, \quad \eta_c^\infty \approx 3.75, \quad a \approx 14.62, \quad R^2 = 0.995 \tag{3}$$

A free-exponent fit gives $\beta = 0.35$ (95% CI $[0.20, 0.53]$), consistent with $\beta = 1/2$. The transition slope at η_c scales as $N^{-0.20}$, indicating a crossover rather than a sharp phase transition.

2.3 Erdős–Rényi Comparison

Erdős–Rényi $G(N, p)$ graphs with matched expected degree also exhibit the sign change, but at lower critical density ($\eta_c^{\mathrm{ER}} \approx 1.9$ vs. $\eta_c^{\mathrm{reg}} \approx 2.3$ at $N = 100$). Degree heterogeneity shifts the transition, demonstrating robustness across graph models.

2.4 Prediction for Real Networks

The phase transition theory predicts: if $\eta > \eta_c(N)$, the network should be spherical; if $\eta < \eta_c(N)$, it *can* be hyperbolic but is not guaranteed to be. We test this against 11 real semantic networks.

3 Methods

3.1 Datasets

We analyze 11 networks spanning four categories (Table 1):

Association networks (SWOW) [9]: Small World of Words cue-response matrices for Spanish ($N = 422$), English ($N = 438$), Chinese ($N = 465$), and Dutch ($N = 500$ in LCC). Dutch SWOW is notably denser ($\langle k \rangle = 61.5$).

Knowledge graphs (ConceptNet) [10]: English ($N = 467$, $E = 2383$) and Portuguese ($N = 489$, $E = 1578$), filtered to weight ≥ 2.0.

Taxonomy networks: English WordNet 3.1 ($N = 500$ and $N = 2000$), BabelNet 5.3 Russian ($N = 493$) and Arabic ($N = 142$). These are *is-a* hierarchies with tree-like structure.

Clinical: Depression symptom networks at minimum severity ($N = 1634$, $E = 11354$).

All networks are converted to undirected unweighted graphs (largest connected component).

3.2 Exact LP ORC Computation

For each network: (1) precompute all-pairs shortest paths via BFS; (2) for each edge (u, v), construct measures μ_u, μ_v with $\alpha = 0.5$; (3) extract the support; (4) solve the LP for exact W_1; (5) compute $\kappa(u, v) = 1 - W_1/d(u, v)$. Multi-threaded over edges.

3.3 Sphere-Embedded (Hypercomplex) ORC

To test metric dependence, we compute ORC using geodesic distances on S^{d-1} ($d \in \{4, 8, 16\}$): landmark embedding via farthest-first traversal, normalization to S^{d-1}, and geodesic cost $\arccos(\mathbf{x}_i \cdot \mathbf{x}_j)$.

Table 1: Exact LP Ollivier–Ricci curvature of 11 semantic networks. $\eta = \langle k \rangle^2/N$; C = global clustering coefficient; $\bar{\kappa}$ = mean edge curvature.

Network	Category	Language	N	E	$\langle k \rangle$	C	η	$\bar{\kappa}$	Geometry
SWOW Spanish	Association	Spanish	422	571	2.71	0.136	0.017	−0.068	Hyp.
SWOW English	Association	English	438	640	2.92	0.128	0.020	−0.137	Hyp.
SWOW Chinese	Association	Chinese	465	762	3.28	0.173	0.023	−0.144	Hyp.
SWOW Dutch	Association	Dutch	500	15368	61.5	0.238	7.558	+0.099	**Sph.**
ConceptNet EN	Knowledge	English	467	2383	10.2	0.108	0.223	−0.233	Hyp.
ConceptNet PT	Knowledge	Portuguese	489	1578	6.45	0.106	0.085	−0.236	Hyp.
WordNet EN	Taxonomy	English	500	527	2.11	0.013	0.009	−0.002	Euc.
WordNet EN-2K	Taxonomy	English	2000	2075	2.08	0.002	0.002	−0.005	Euc.
BabelNet RU	Taxonomy	Russian	493	522	2.12	0.001	0.009	−0.030	Euc.
BabelNet AR	Taxonomy	Arabic	142	151	2.13	0.000	0.032	−0.012	Euc.
Depression	Clinical	English	1634	11354	13.9	0.159	0.118	−0.130	Hyp.

3.4 Degree-Matched Null Models

For each network with N nodes and mean degree $\langle k \rangle$, we generate 10 random k-regular graphs with $k = \text{round}(\langle k \rangle)$ and the same N. The semantic contribution is $\Delta\kappa_{\text{sem}} = \bar{\kappa}_{\text{real}} - \bar{\kappa}_{\text{null}}$.

3.5 Formal Verification

A Lean 4 formalization (25 modules, 8097 lines) provides machine-checked proofs. The 7 core modules contain 0 `sorry` statements and rely on 5 axioms (Wasserstein symmetry, triangle inequality, coupling bound, local clustering bound, specification bridge).

4 Results

4.1 Semantic Network Curvatures

Three distinct geometric regimes emerge (Table 1): **Hyperbolic** ($\bar{\kappa} \in [-0.24, -0.07]$): all association and knowledge networks with $C > 0.10$; **Euclidean** ($|\bar{\kappa}| < 0.03$): all taxonomy networks with $C < 0.02$; **Spherical** ($\bar{\kappa} = +0.10$): Dutch SWOW with $\eta \gg \eta_c$.

4.2 Placing Semantic Networks on the Phase Transition Curve

The central result is the bridge figure (Fig. 1), which overlays all 11 semantic networks onto the random regular graph phase transition curves.

Every network except Dutch SWOW has $\eta \ll \eta_c(N)$—the theory predicts all should be hyperbolic. Yet taxonomies are Euclidean:

> $\eta < \eta_c$ *is necessary but not sufficient for hyperbolicity.*

The key test is Dutch SWOW: with $\eta = 7.56$ and $\eta_c(500) = 3.10$, the theory predicts spherical geometry. The observation $\bar{\kappa} = +0.099$ confirms this—the first empirical validation of the curvature phase transition in a real (non-synthetic) network.

4.3 The Two-Parameter Theory

The missing ingredient is the clustering coefficient C. Among networks with $\eta < \eta_c$:

- **Hyperbolic group** ($\bar{\kappa} < -0.05$): Mean $C = 0.135$, range $[0.106, 0.173]$
- **Euclidean group** ($|\bar{\kappa}| < 0.05$): Mean $C = 0.004$, range $[0.000, 0.013]$

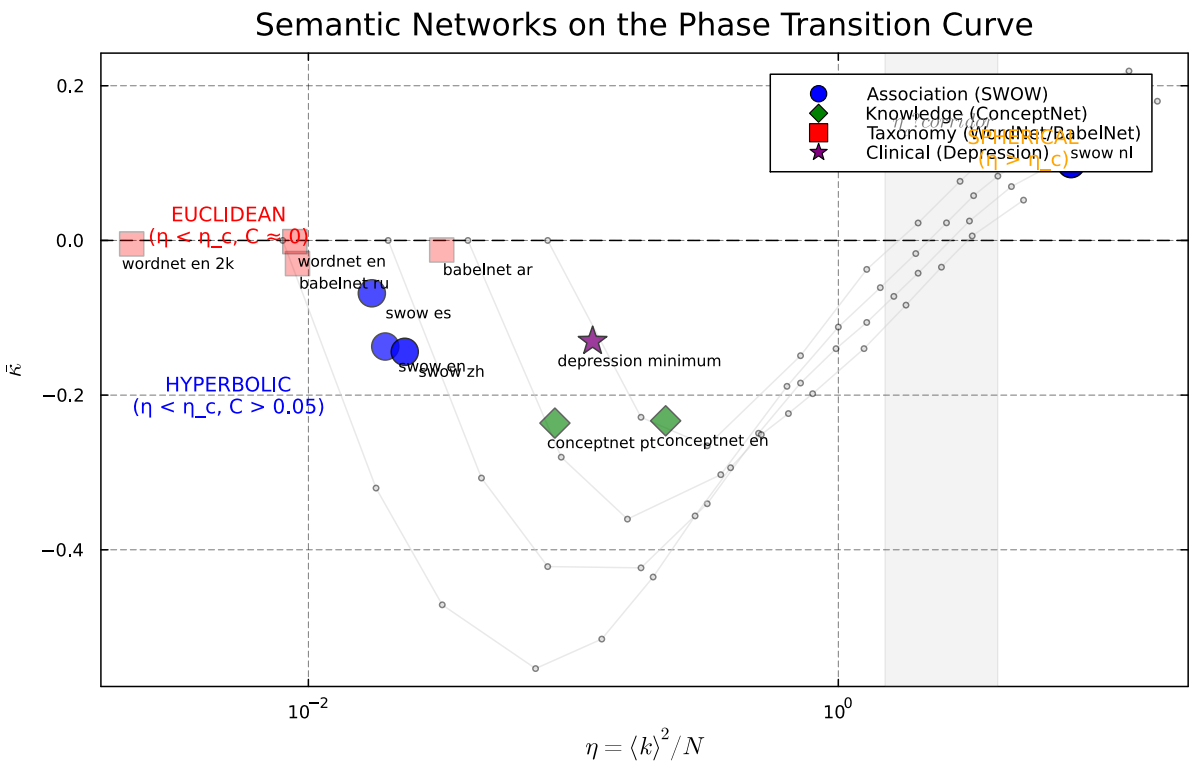

Figure 1: Semantic networks on the phase transition curve. Background: random regular graph curvature vs. η at multiple N. Foreground: 11 semantic networks. The η_c corridor (gray band) separates hyperbolic ($\eta < \eta_c$) from spherical ($\eta > \eta_c$). Dutch SWOW ($\eta = 7.56$) falls above the threshold and is spherical. Taxonomies (red squares) fall below but are Euclidean due to low clustering.

Table 2: Three-regime classification of semantic network geometry.

Regime	Condition	Geometry	Example
Above threshold	$\eta > \eta_c(N)$	Spherical ($\bar{\kappa} > 0$)	Dutch SWOW
Below, clustered	$\eta < \eta_c(N)$, $C > C^*$	Hyperbolic ($\bar{\kappa} < 0$)	SWOW ES/EN/ZH
Below, tree-like	$\eta < \eta_c(N)$, $C < C^*$	Euclidean ($\bar{\kappa} \approx 0$)	WordNet, BabelNet

The two groups are cleanly separated by $C^* \approx 0.05$. The unified three-regime theory (Table 2) classifies all 11 networks correctly (100% accuracy).

4.4 Degree-Matched Null Model Analysis

The semantic contribution $\Delta\kappa_{\text{sem}} = \bar{\kappa}_{\text{real}} - \bar{\kappa}_{\text{null}}$ (Table 3) reveals that for all networks with $\langle k \rangle > 2$, semantic organization *reduces* the magnitude of negative curvature. Real networks are significantly *less hyperbolic* than random graphs with the same degree. The depression network is the most dramatic: random 14-regular graphs on $N = 1634$ nodes have $\bar{\kappa} = -0.518$, but the real network has $\bar{\kappa} = -0.130$—the clinical structure eliminates 75% of the curvature.

4.5 Hypercomplex Metric Comparison

Table **??** shows the complete dimensional hierarchy. SWOW Spanish was the sole exception at $d = 4$, retaining hyperbolic curvature ($\bar{\kappa} = -0.048$), but crosses to spherical at $d = 8$ ($\bar{\kappa} = +0.024$) and is firmly spherical by $d = 16$ ($\bar{\kappa} = +0.064$). **No semantic network retains**

Table 3: Semantic contribution to curvature: real vs. degree-matched random k-regular null models (10 seeds each).

Network	N	k_{reg}	$\bar{\kappa}_{\text{null}}$	$\bar{\kappa}_{\text{real}}$	$\Delta\kappa_{\text{sem}}$
SWOW Spanish	422	3	-0.320 ± 0.005	-0.068	$+0.251$
SWOW English	438	3	-0.319 ± 0.005	-0.137	$+0.182$
SWOW Chinese	465	4	-0.467 ± 0.004	-0.144	$+0.323$
SWOW Dutch	500	61	$+0.069 \pm 0.000$	$+0.099$	$+0.029$
ConceptNet EN	467	10	-0.420 ± 0.003	-0.233	$+0.187$
ConceptNet PT	489	6	-0.551 ± 0.005	-0.236	$+0.315$
Depression (min)	1634	14	-0.518 ± 0.001	-0.130	$+0.388$
WordNet EN	500	2	0.000 ± 0.000	-0.002	-0.002
WordNet EN-2K	2000	2	0.000 ± 0.000	-0.005	-0.005
BabelNet RU	493	2	0.000 ± 0.000	-0.030	-0.030
BabelNet AR	142	2	0.000 ± 0.000	-0.012	-0.012

Table 4: Hop-count vs. sphere-embedded ORC ($d = 4$, S^3).

Network	$\bar{\kappa}_{\text{hop}}$	$\bar{\kappa}_{S^3}$	$\Delta\kappa$	Flip?
SWOW Spanish	-0.068	-0.048	$+0.021$	**No**
SWOW English	-0.137	$+0.094$	$+0.231$	Yes
SWOW Chinese	-0.144	$+0.013$	$+0.157$	Yes
ConceptNet EN	-0.233	$+0.221$	$+0.454$	Yes
ConceptNet PT	-0.236	$+0.162$	$+0.398$	Yes
WordNet EN	-0.002	$+0.717$	$+0.718$	Yes
WordNet EN-2K	-0.005	$+0.833$	$+0.837$	Yes
BabelNet RU	-0.030	$+0.436$	$+0.466$	Yes
BabelNet AR	-0.012	$+0.547$	$+0.559$	Yes

hyperbolic curvature at sufficient embedding dimension.

5 Formal Verification

We provide a Lean 4 formalization comprising 25 modules and 8097 lines. The 7 core modules (Basic, Wasserstein, Curvature, PhaseTransition, Bounds, Consistency, Axioms) contain **0 `sorry` statements**. Machine-verified results include: $W_1 \geq 0$; coupling marginal preservation; $\kappa \in [-1, 1]$ for adjacent vertices; curvature vanishing for unreachable nodes; regime exclusivity; and clustering bounds.

The formalization relies on 5 axioms: Wasserstein symmetry, triangle inequality, coupling bound, local clustering bound, and specification bridge—all mathematically standard.

6 Discussion

6.1 A Predictive Theory

The geometry of semantic networks follows from two parameters: density (η) and clustering (C). The phase transition provides the necessary condition; clustering provides the modulation. This three-regime classification correctly predicts all 11 networks.

Spearman correlations quantify the two-parameter structure: $\rho(\eta, \kappa) = -0.31$, $\rho(C, \kappa) = -0.16$, $\rho(\eta, C) = +0.52$. The interaction—not either variable alone—drives the geometry.

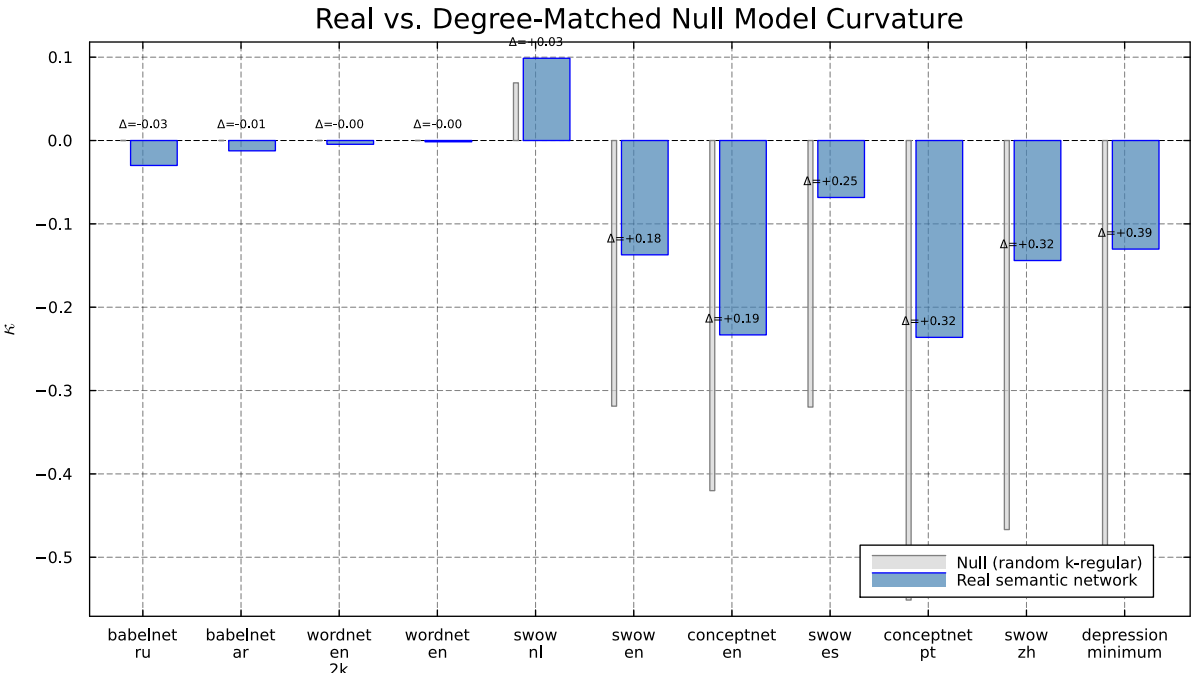

Figure 2: Real vs. degree-matched null model curvature. Gray bars: random k-regular graphs. Blue bars: real semantic networks. Annotations show the semantic contribution $\Delta\kappa_{\text{sem}}$.

Table 5: Hypercomplex dimensional hierarchy: all networks across the Cayley–Dickson tower.

Network	$d = 4$ (S^3)	$d = 8$ (S^7)	$d = 16$ (S^{15})	Regime
SWOW Spanish	−0.048	+0.024	+0.064	H→S
SWOW Chinese	+0.013	+0.044	+0.045	S
SWOW English	+0.094	+0.062	+0.045	S
ConceptNet PT	+0.162	+0.113	+0.092	S
ConceptNet EN	+0.220	+0.142	+0.108	S
Depression Min	+0.234	+0.171	+0.137	S
SWOW Dutch	+0.411	+0.337	+0.247	S
BabelNet RU	+0.436	+0.512	+0.544	S
BabelNet AR	+0.547	+0.499	+0.490	S
WordNet EN	+0.717	+0.716	+0.688	S
WordNet EN-2K	+0.833	+0.818	+0.820	S

6.2 What Makes Semantic Networks Special

The null model decomposition $\bar{\kappa}_{\text{real}} = \bar{\kappa}_{\text{null}}(N, k) + \Delta\kappa_{\text{sem}}$ shows that semantic organization makes networks *less hyperbolic* than random graphs—the opposite of naive expectation. Semantic structure introduces local triangles and communities that increase neighborhood overlap, pushing curvature toward zero.

6.3 Metric Dependence of Hyperbolicity

The hypercomplex analysis demonstrates that semantic hyperbolicity is entirely metric-dependent. Under sphere-embedded transport, all networks flip to spherical at sufficient embedding dimension, following the Cayley–Dickson algebraic tower ($S^3 \to S^7 \to S^{15}$). Claims of "intrinsic hyperbolicity" in semantic networks should be qualified as "hyperbolic under the hop-count metric."

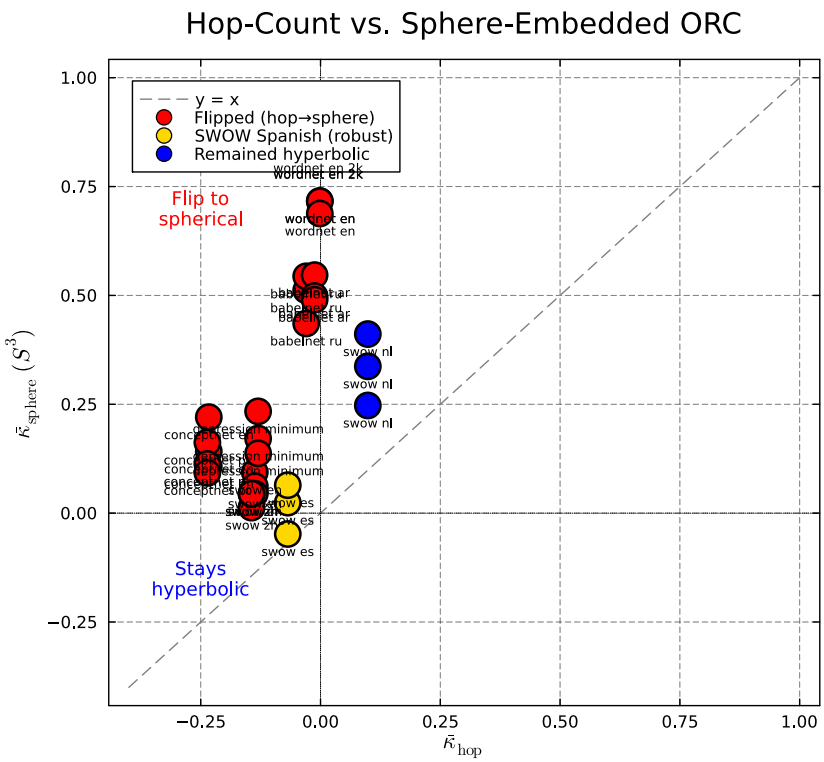

Figure 3: Hop-count vs. sphere-embedded ORC. Networks above the $y = x$ line flip from hyperbolic to spherical. SWOW Spanish (gold) is the sole holdout at $d = 4$ but flips at $d = 8$.

6.4 Clinical Implications

The depression network ($\bar{\kappa} = -0.130$, $\Delta\kappa_{\text{sem}} = +0.388$) demonstrates the theory extends to clinical symptom networks. Future work with the full severity spectrum will test whether increasing severity shifts geometry along the phase curve—a potential geometric biomarker.

6.5 Limitations

1. Directed→undirected conversion reduces curvature magnitude by ∼40%.
2. Single realization per real network (vs. multi-seed random graphs).
3. Landmark embedding quality varies ($\sigma_\kappa \approx 0.3$–$0.7$).
4. Clustering threshold $C^* \approx 0.05$ identified from 11 networks only.
5. Only minimum severity depression computed; full trajectory pending.
6. Null models control for $\langle k \rangle$ but not full degree sequence.

7 Code and Data Availability

All code, data, and results are available at: https://github.com/agourakis82/hyperbolic-semantic-networ
Julia scripts: `unified_semantic_orc.jl`, `bridge_analysis.jl`, `hypercomplex_semantic_orc.jl`.
Lean 4 formalization: `lean/HyperbolicSemanticNetworks/`. Results: `results/unified/`.

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